

Solving initial value problems (IVP) for ordinary differential equations (ODE)

Cauchy problem

Given $I = [0, T] \subseteq \mathbb{R}$ $f: I \times \mathbb{R} \Rightarrow \mathbb{R}$

find $y: I \rightarrow \mathbb{R}$ s.t.

Differential $\rightarrow \begin{cases} y'(t) = f(t, y) & 0 \leq t \leq T \end{cases} \quad (1)$

Initial condition $\rightarrow \begin{cases} y(0) = y_0 \end{cases}$

Theorem (Cauchy)

- Let f is continuous w.r.t. both arguments
- Let f is Lipschitz-continuous w.r.t. the

2nd argument

$$|f(t, y_1) - f(t, y_2)| \leq \textcolor{red}{L} |y_1 - y_2| \quad \forall t \in I \\ \forall y_1, y_2 \in \mathbb{R}$$

$\swarrow$
Lipschitz constant

- Then the solution to the Cauchy problem (1)
- exists
- C^1
- Depends continuously on y_0

Example

$$\begin{cases} y'(t) = 3y(t) - 3t & 0 < t \leq 2 \\ y(0) = 1 \end{cases}$$

$$f(t, y) = 3y - 3t$$

$$|f(t, y_1) - f(t, y_2)| = |3y_1 - 3t - (3y_2 - 3t)|$$

$$= |3y_1 - 3y_2| = \underbrace{(3)}_L |y_1 - y_2|$$

$$\exists! \text{ sol. : } y(t) = \frac{2}{3} e^{3t} + t + \frac{1}{3}$$

Numerical integration of ODE problems

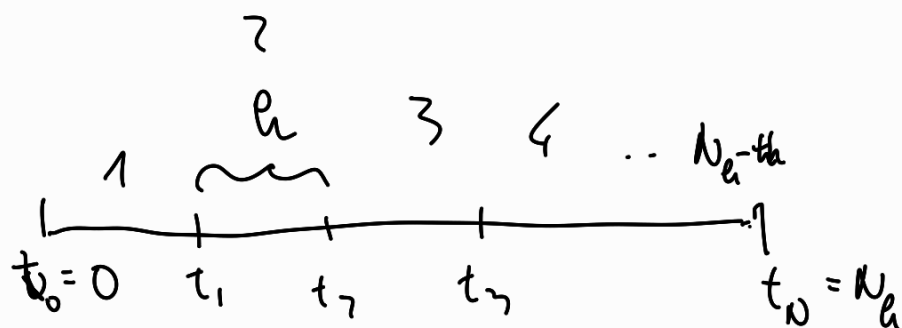
► $I \equiv [0, T]$ $T < +\infty$

► Subdivide I in N_h subintervals

Using $N_h + 1$ equally spaced points

$$0 = t_0 < t_1 < t_2 < \dots < t_{N_h} = T$$

$$t_i - t_{i-1} = h \quad \forall i = 1 \dots N_h$$



► Find $N_h + 1$ values $\{u_0, u_1, \dots, u_{N_h}\}$

that "approximate" the values $y(t_0), y(t_1), \dots$

$y(T)$ of the solution to the Cauchy problem

$$y_i = y(t_i)$$

$$f_i = f(t_i, y_i)$$

$$d_h^+ y(t) - y'(t) \approx \mathcal{O}(h)$$

$$d_h^- y(t) - y'(t) \approx \mathcal{O}(h)$$

$$d_h^c y(t) - y'(t) \approx \mathcal{O}(h^2)$$

$$d_h^+ y(t) = \frac{y(t+h) - y(t)}{h}$$

$$d_h^- y(t) = \frac{y(t) - y(t-h)}{h}$$

$$d_h^c y(t) = \frac{y(t+h) - y(t-h)}{2h}$$

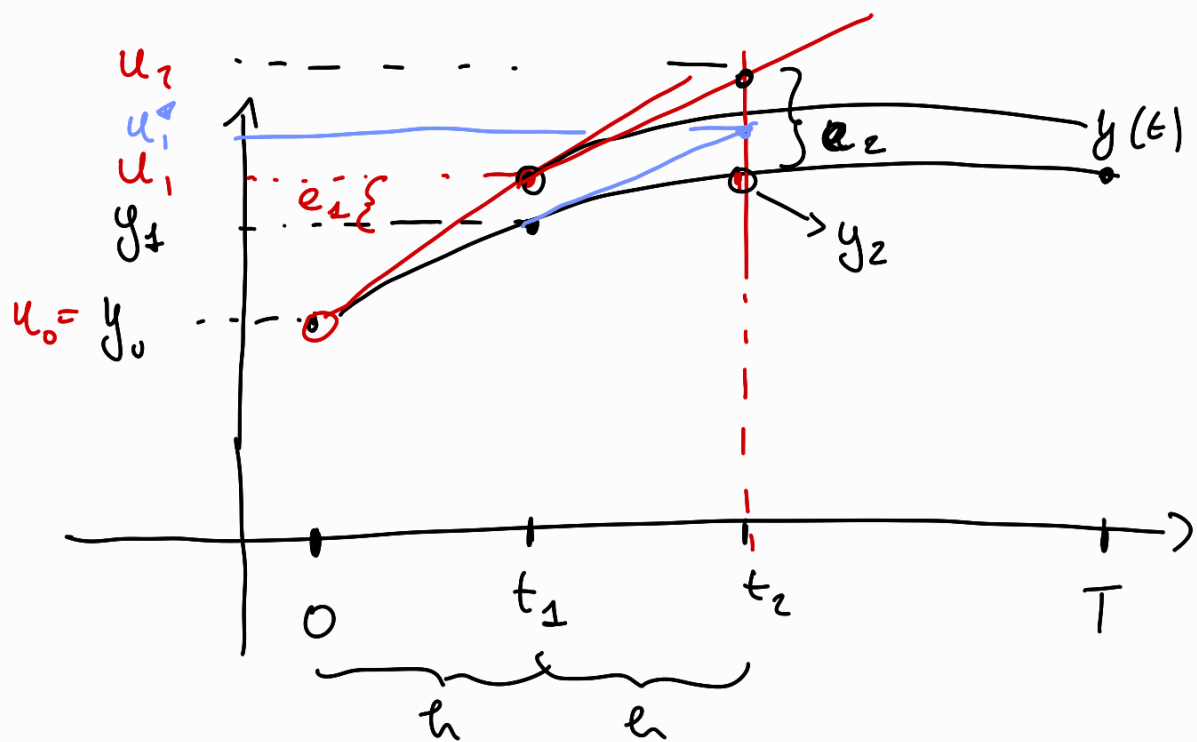
$$\begin{cases} y'(t) = f(t, y(t)) & t \in I \\ y(0) = y_0 \end{cases} \Rightarrow \begin{cases} \frac{u_{n+1} - u_n}{h} = f(t_n, u_n) \\ u_0 = y_0 \end{cases}$$

$$\begin{cases} u_{n+1} = u_n + h f(t_n, u_n) \\ u_0 = y_0 \end{cases}$$

recursive formula
to generate the sequence
 $u_1, u_2, u_3, \dots, u_N$

Forward Euler's method

(FWE)



Backward Euler's Method (BWE)

$$\begin{cases} y'(t) = f(t, y) & t \in I \\ y(0) = y_0 \end{cases} \Rightarrow \begin{cases} \frac{u^n - u^{n-1}}{h} = f(t_n, u_n) \\ u_0 = y_0 \end{cases}$$

$$\underline{u_n} = u_{n-1} + h f(t_n, \underline{u_n}) \quad \text{implicit method}$$

$$u_0 = y_0$$

At each step, knowing u_{n-1} I need to solve a (non-linear) problem of the form

$$\text{find } u_n \text{ s.t. } g(u_n) = 0$$

$$g(u_n) = u_n - u_{n-1} - h f(t_n, u_n)$$

Explicit one-step methods

$$u_{n+1} = u_n + h \Phi_a(u_n) \quad \cancel{u_{n-1}} \quad \cancel{u_{n-2}}$$

Implicit one-step methods

$$u_{n+1} = u_n + h \Phi_a(u_{n+1}, u_n) \quad \cancel{u_{n-1}} \quad \cancel{u_{n-2}}$$

Local truncation error (for explicit one-step methods)

$$u_{n+1} = u_n + h \Phi_a(t_n, u_n)$$

$$u_{n+1}^* = y_n + h \Phi_a(t_n, y_n)$$

$$e_{n+1} = u_{n+1} - y_{n+1} = \underbrace{(u_{n+1} - y_{n+1})}_{h \tau_n(e)} + \underbrace{(u_{n+1} - u_{n+1}^*)}$$

$$\tau_n(e) = \frac{u_{n+1}^* - y_{n+1}}{h}$$

local truncation error

$$\tau(e) = \max_n |\tau_n(e)|$$

global truncation error

An explicit one-step method is said to be consistent

if $\lim_{h \rightarrow 0} T_n(h) = 0$

It is said to be order p if

$$T(a) = \mathcal{O}(a^p)$$

A method is convergent if $\lim_{h \rightarrow 0} e_n(h) = 0 \quad \forall n$

FWE

$$- \quad u_{n+1}^* = y_n + h f(t_n, y_n)$$

$$\frac{u_{n+1}^* - y_n}{h} = f(t_n, y_n) = y'(t_n)$$

$$- \quad y_{n+1} = y_n + h y'(t_n) + \frac{h^2}{2} y''(\xi)$$

$$T_n(h) = \frac{u_{n+1}^* - y_{n+1}}{h} = \frac{h}{2} y''(\xi) \xrightarrow{h} 0$$

if y'' is bounded

$$T(a) \leq M a / 2 \quad \mathcal{O}(a)$$

$$e_n = \underbrace{(u_n - u_n^*)}_{?} + \underbrace{(u_n^* - y_n)}_{h\tau_n}$$

$$|e_n| \leq \underline{|u_n - u_n^*|} + |u_n^* - y_n|$$

$$\begin{aligned} |u_n - u_n^*| &= |u_{n-1} + h f(t_{n-1}, u_{n-1}) - y_{n-1} - h f(t_{n-1}, y_{n-1})| \\ &= |e_{n-1} + h (f(t_{n-1}, u_{n-1}) - f(t_{n-1}, y_{n-1}))| \\ &\leq |e_{n-1}| + hL |u_{n-1} - y_{n-1}| \\ &\leq |e_{n-1}| + hL |e_{n-1}| \\ &= (1 + hL) |e_{n-1}| \end{aligned}$$

$$|e_n| \leq h|\tau_n(a)| + |e_{n-1}|(1 + hL)$$

$$\leq \frac{e^{L\tau}}{\tau} \tau(a)$$

Absolute stability for ODE integrators

One step explicit methods for IVP for ODEs

$$u_{n+1} = u_n + h \Phi(t_n, u_n)$$
$$u_0 = y_0$$

One step implicit methods

$$u_{n+1} = u_n + h \phi_a(t_{n+1}, \underline{u_{n+1}}, t_n, \underline{u_n})$$

$$u_0 = y_0$$

FWE

$$\begin{cases} u_{n+1} = u_n + h f(t_n, u_n) \\ u_0 = y_0 \end{cases}$$

$$\begin{cases} y' = f(t, y) \\ y(0) = y_0 \end{cases}$$

order 1 explicit

BWE

$$u_{n+1} = u_n + h f(t_{n+1}, u_{n+1})$$

$$u_0 = y_0$$

order 1 implicit

CN (Crank Nicholson)

$$u_{n+1} = u_n + \frac{h}{2} \{ f(t_n, u_n) + f(t_{n+1}, \underline{u_{n+1}}) \}$$

$$u_0 = y_0$$

order 2 implicit

Trapezoid method

$$y_{t_{n+1}} - y_{t_n} = \int_{t_n}^{t_{n+1}} y' = \int_{t_n}^{t_{n+1}} f(t, y(t)) \approx \frac{t_{n+1} - t_n}{2} (f(t_n, y(t_n)) + f(t_{n+1}, y(t_{n+1})))$$

Heun's method

C.N. $u_{n+1} = u_n + \frac{h}{2} \left\{ f(t_n, u_n) + f(t_{n+1}, \textcircled{u_{n+1}}) \right\}$

implicit
↓

$$u_{n+1}^* = u_n + h f(t_n, u_n) \quad \text{FWE}$$

$$\begin{cases} u_{n+1} = u_n + \frac{h}{2} \left\{ f(t_n, u_n) + f(t_{n+1}, u_{n+1}^*) \right\} \\ u_0 = y_0 \end{cases}$$

order 2 explicit

Absolute Stability

Model problem:

$$\begin{cases} y' = \lambda y & 0 < t \leq T \longrightarrow \infty \\ y(0) = 1 \end{cases}$$

$$|y(t)| = |e^{\lambda t}| \begin{cases} \xrightarrow{t \rightarrow \infty} 0 & \text{if } \operatorname{Re}(\lambda) < 0 \\ \xrightarrow{t \rightarrow \infty} 1 & \text{if } \operatorname{Re}(\lambda) = 0 \\ \xrightarrow{t \rightarrow \infty} \infty & \text{if } \operatorname{Re}(\lambda) > 0 \end{cases}$$

FWE

$$u_{n+1} = u_n + h\lambda u_n$$

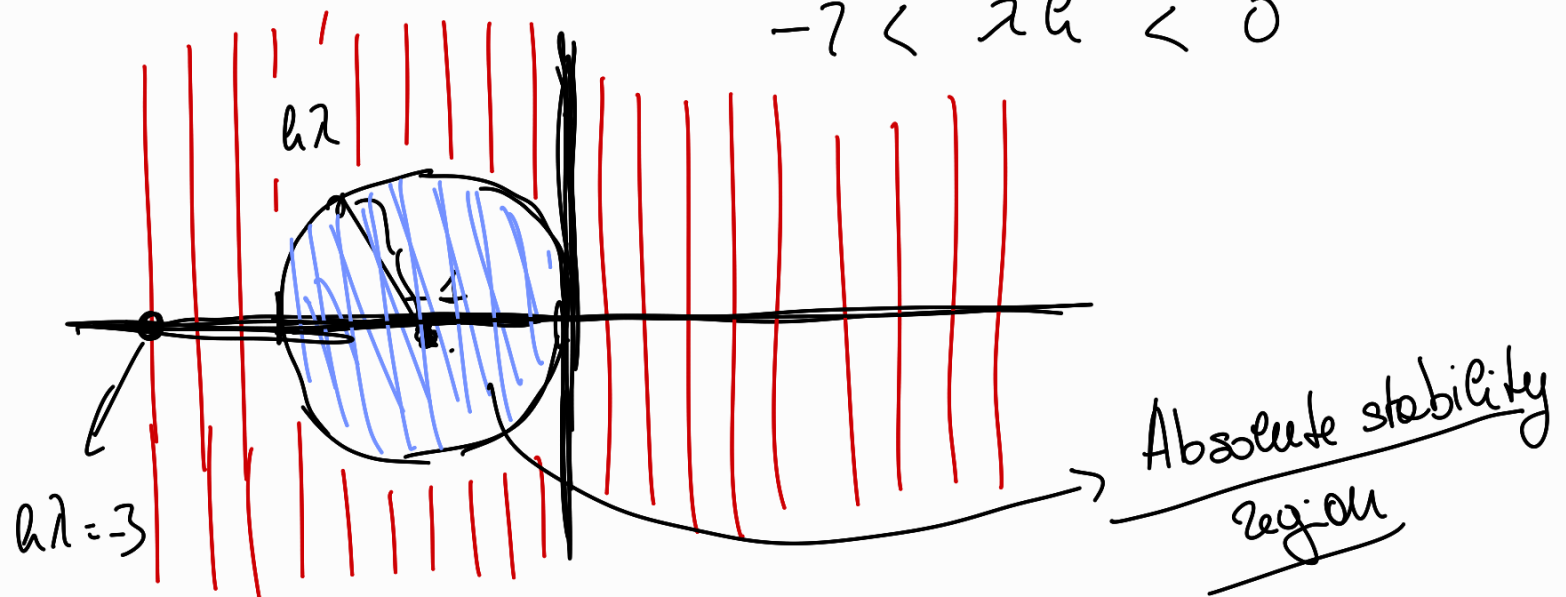
$$= u_n (1 + \lambda h)$$

$$|u_{n+1}| < |u_n| \quad |1 + \lambda h| < 1$$

$$\lambda \in \mathbb{R}$$

$$-1 < 1 + \lambda h < 1$$

$$-2 < \lambda h < 0$$



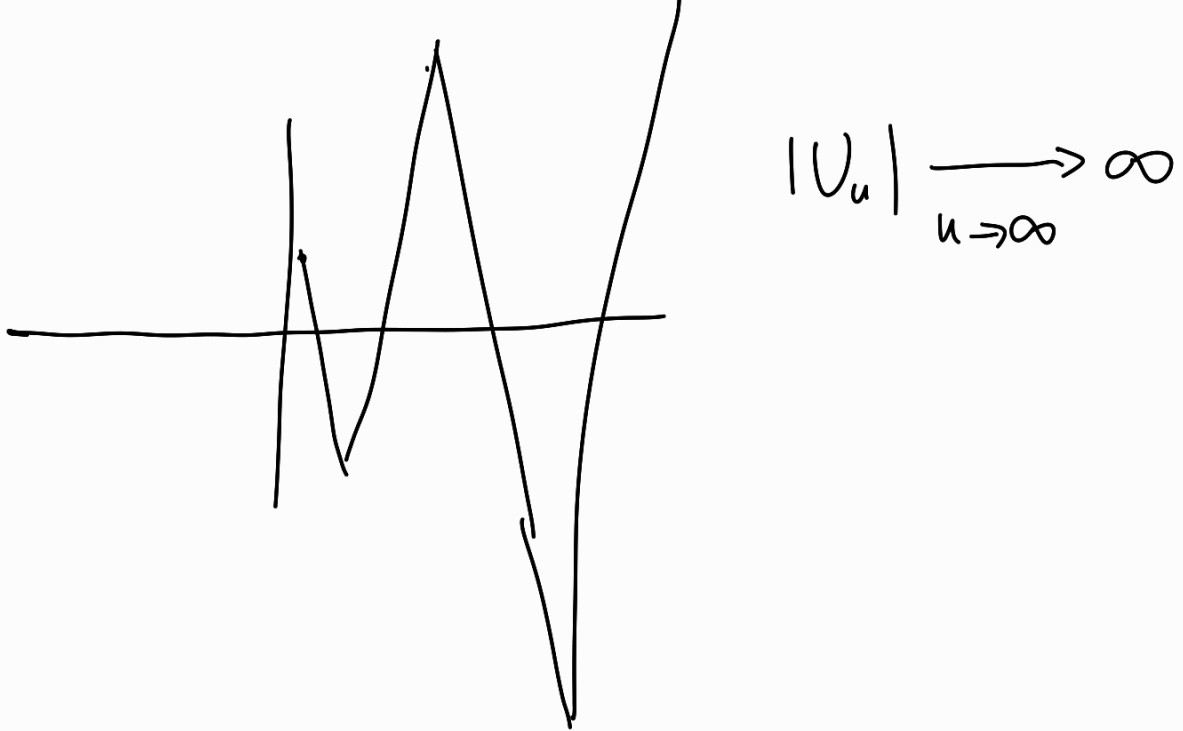
$$\lambda = -3$$

$$h = 1$$

$$u_{n+1} = u_n + 1(-3)u_n$$

$$u_{n+1} = u_n (-2)$$

$$u_0 = 1 \quad u_1 = -2 \quad u_2 = 4 \quad u_3 = -8 \dots$$

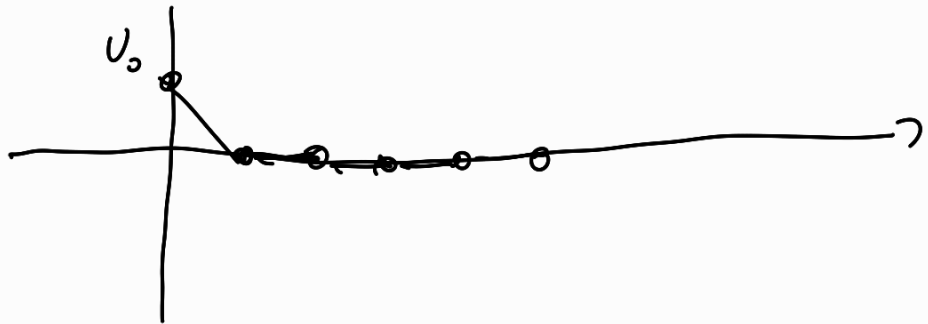


$$\lambda = -3$$

$$u_{n+1} = u_n + \frac{1}{3}(-3) u_n$$

$$h = \frac{1}{3}$$

$$u_{n+1} = u_n \cdot (1 - 1) = 0$$



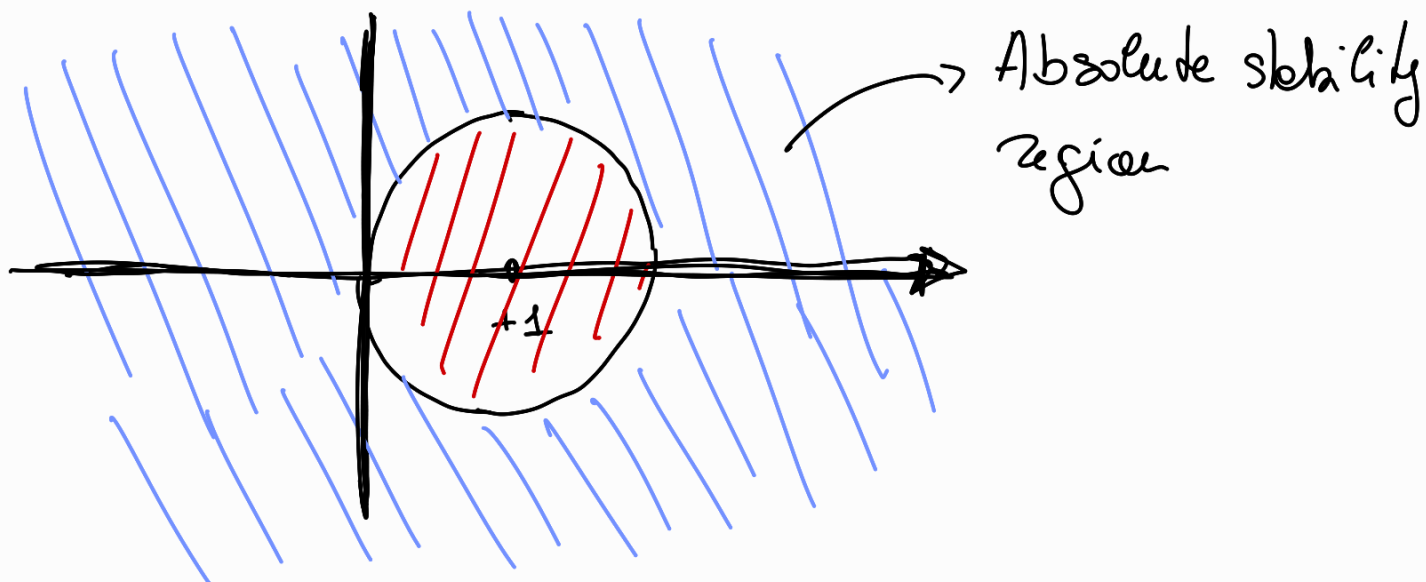
BWE

$$u_{n+1} = u_n + h\lambda u_{n+1}$$

$$u_{n+1} = u_n \left(\frac{1}{1 - h\lambda} \right)$$

$$|u_{n+1}| < |u_n| \quad \left| \frac{1}{1 - h\lambda} \right| < 1$$

$$1 < |1 - \lambda \tau|$$



unconditionally absolutely stable

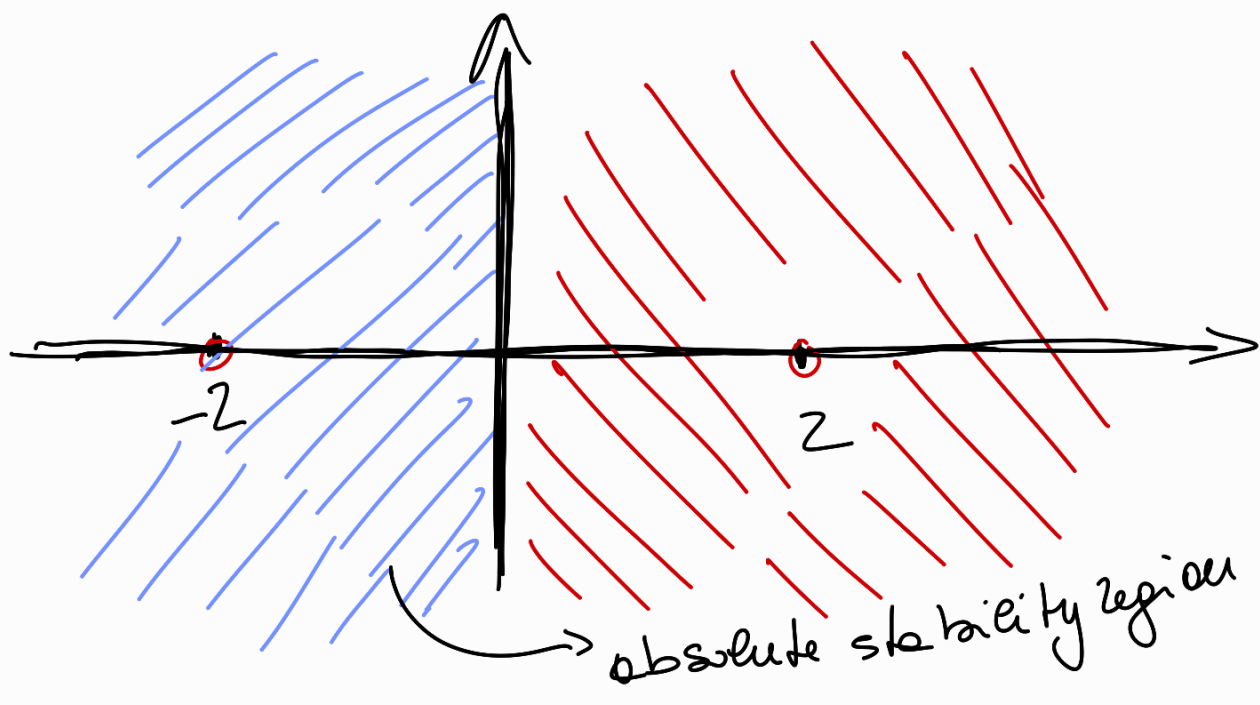
Crank Nicolson

$$u_{n+1} = u_n + \frac{\lambda \tau}{2} (u_n + u_{n+1})$$

$$u_{n+1} \left(1 - \frac{\lambda \tau}{2} \right) = u_n \left(1 + \frac{\lambda \tau}{2} \right)$$

$$u_{n+1} = u_n \frac{1 + \frac{\lambda \tau}{2}}{1 - \frac{\lambda \tau}{2}}$$

$$\left| \frac{1 + \frac{\lambda \tau}{2}}{1 - \frac{\lambda \tau}{2}} \right| < 1$$



unconditionally absolutely stable method

$$\text{if } \operatorname{Re}(\lambda) = 0 \quad \|u_n\| = 1 \quad \forall n$$

Theorem There is no unconditionally absolutely
stable explicit method

Systems of ODEs

$$\begin{cases} y'(t) = f(t, y(t)) & 0 < t \leq T \\ y(0) = y_0 \end{cases}$$

$$f: [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d \quad y: [0, T] \rightarrow \mathbb{R}^d$$

- The Cauchy theorem can be trivially extended to the vector case

$$\|f(t, y_1) - f(t, y_2)\| \leq L \|y_1 - y_2\|$$

$$y'' = f(y, y') \Leftrightarrow \begin{cases} z(t) = y'(t) \\ z'(t) = f(y, z) \end{cases}$$

$$\frac{d}{dt} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} z(t) \\ f(y(t), z(t)) \end{bmatrix}$$

Any IVP of order $p > 1$
can be reduced to a
system of order 1 ODEs

$$m \ddot{x} = -\lambda x$$

harmonic oscillator

$$\begin{cases} \ddot{x} = -\frac{\lambda}{m} x \\ m \dot{x} = -\lambda x \end{cases}$$

Linear autonomous system of ODEs

$$\begin{cases} \underline{y}' = \underline{A} \underline{y}(t) & 0 < t < T \\ \underline{y}(0) = \underline{y}_0 \end{cases}$$

Assume the A be symmetric and diagonalizable

$$A = U \Lambda U^T \quad U^T U = I$$

$$\Lambda = \text{diag} \{ [\lambda_1, \lambda_2, \lambda_3, \dots] \}$$

$\{\lambda_i\}$ eigenvalues of A

$$y' = Ay = U \Lambda U^T y.$$

$$U^T y' = \Lambda U^T y \quad z = U^T y$$

$$z' = \Lambda z$$

$$\boxed{z_1' = \lambda_1 z_1}$$

$$z_2' = \lambda_2 z_2$$

$$z_3' = \lambda_3 z_3$$

~ . . . ~

"Stiff" problems

$$\left\| \frac{\lambda_{\max}}{\lambda_{\min}} \right\| \gg 1$$

$h\lambda_{\max}$ and $h\lambda_{\min}$

must be within the ASR.

to satisfy both I may need a very small h